Principles of Evolution

Chapter 7

Section 1: Foundations

Clearing common myths and identifying core biological principles.

Addressing Common Myths

"Just a Theory"

In science, a Theory is the highest level of understanding—a well-substantiated explanation supported by facts.

"Individuals Evolve"

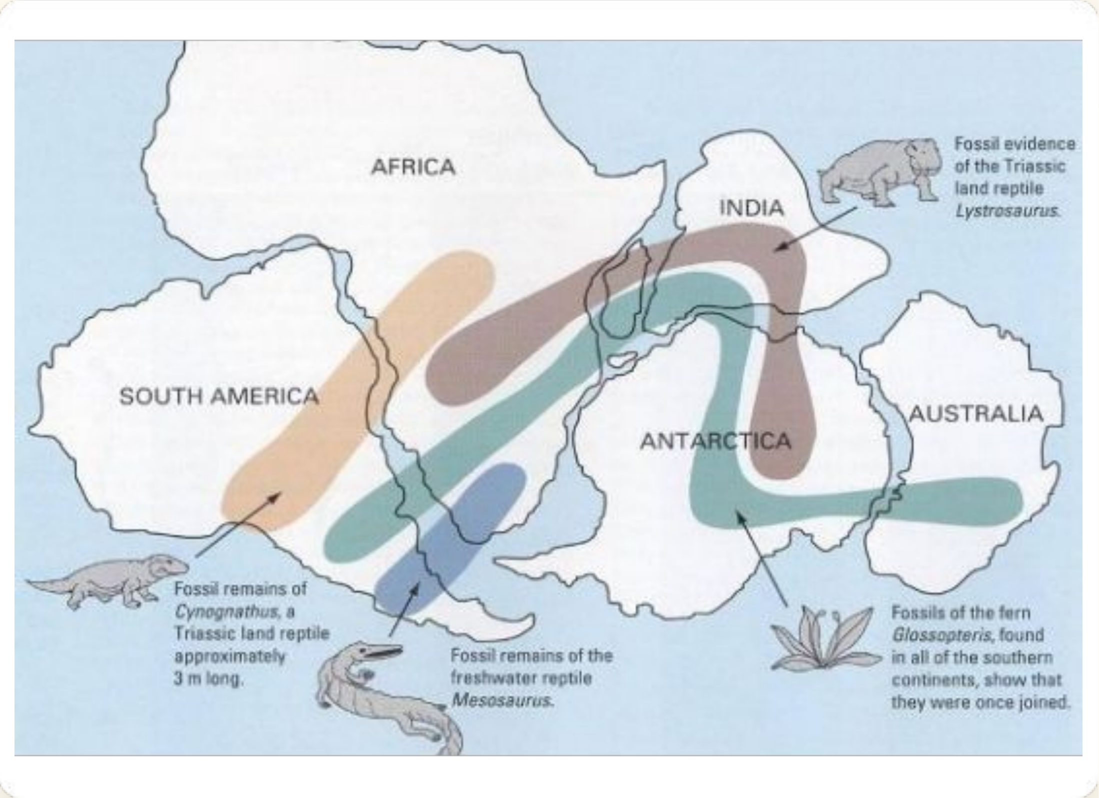
Individuals do not evolve; they live or die. **Populations** evolve as allele frequencies change over time.

"Evolution is Random"

Mutations are random typos, but **Natural Selection** is a non-random environmental filter.

Darwin's Core Principles

- **Variation:** Natural genetic differences exist in every population.
- **Overproduction:** More offspring are born than can possibly survive.
- **Competition:** Resources are limited, leading to a "Struggle for Existence."
- **Differential Survival:** Those with advantageous traits reproduce more successfully.
- **Descent with Modification:** Modern organisms come from ancient ancestors.

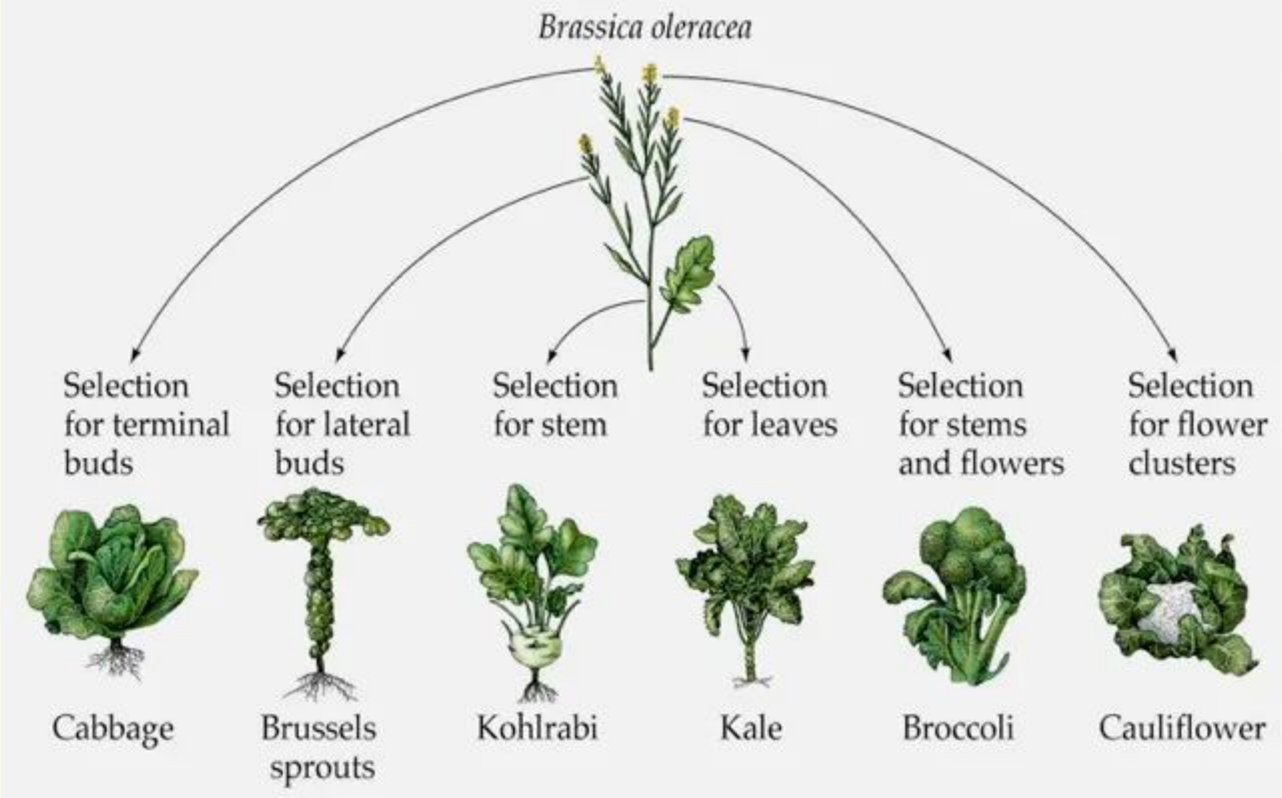


Artificial Selection

Human-Driven Change

Nature isn't the only selective force. Humans have shaped species for millennia.

- **Dogs:** All breeds from Great Danes to Chihuahuas descended from the Gray Wolf.
- **Agriculture:** Wild mustard was selectively bred into kale, broccoli, and cabbage.
- **Saskatchewan:** Canola (Low Acid Canadian Oil) was created through selective breeding of rapeseed.



Universal Common Ancestor (LUCA)

Genetic Code

The language of DNA (A,T,C,G) is identical across all living things on Earth.

ATP Energy

Every known organism uses Adenosine Triphosphate as its primary cellular energy currency.

Wâhkôhtowin

Indigenous kinship teaches "All My Relations"—a biological fact confirmed by common ancestry.

Timeline of Evolutionary Theory

Scientist	Major Contribution	Key Idea
Lamarck	Acquired Characteristics	Species change, but mechanism (stretching necks) was incorrect.
Malthus	Struggle for Survival	Economist who noted populations outgrow food supply.
Darwin/Wallace	Natural Selection	Mechanism: Differential survival and reproduction.
Wegener	Continental Drift	Geology explains why fossils are separated by oceans.
Gould	Punctuated Equilibrium	Brief bursts of rapid change following catastrophes.

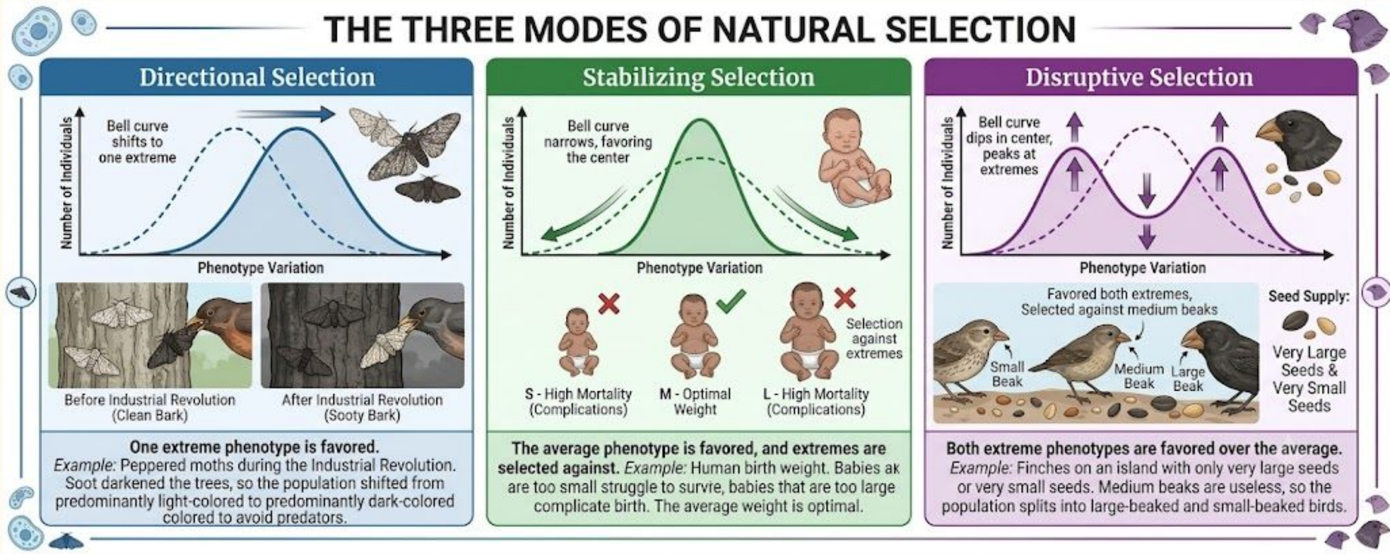
Darwin's Key Observations

Three Patterns of Diversity

- **Species Vary Globally:** Similar birds (Rheas, Ostriches, Emus) in similar habitats worldwide.
- **Species Vary Locally:** Galapagos finches adapted unique beak shapes for specific island foods.
- **Species Vary Over Time:** Ancient Glyptodon fossils look like massive versions of modern armadillos.

Modes of Natural Selection

- **Directional Selection:** Shifts toward one extreme (e.g., Peppered moths turning dark).
- **Stabilizing Selection:** Favors the average, selects against extremes (e.g., Human birth weight).
- **Disruptive Selection:** Favors both extremes, selects against the average (e.g., Large and small beak sizes).



Note: Selection acts on the individual, but the population evolves.

The Path to Speciation

Geographic Isolation

Physical barriers (rivers, canyons) split populations until they interbreed no longer.

Temporal Isolation

Populations reproduce at different times (different days or seasons).

Behavioural Isolation

Differences in courtship rituals or mating songs prevent cross-breeding.

Structural Evidence

- **Homologous Structures:** Identical bone patterns (human arm, bat wing) suggest common ancestry.
- **Vestigial Structures:** "Leftover" parts with no modern function (whale hip bones, human tailbone).
- **Comparative Embryology:** Vertebrate embryos show temporary gill slits and tail buds.